

the end of the timestep, and (iii) there is a single collision within the timestep. [The case of additional collisions within the step could be handled as in (iii).] In cases (i) and (ii) it is clear that the error accumulating in a single step will be third order in the stepsize, since the Verlet method has local error of order three (since it is a second order method).

Now consider case (iii). The contact happens when the particles are separated by a distance 1, at which point $\mathbf{u}_\perp = \mathbf{q}$. We assume the time of contact is τ_c .

$$\begin{aligned}\bar{\mathbf{P}} &:= \mathbf{p} - \frac{h}{2} \nabla U(\mathbf{q}), & \mathbf{q}_c &:= \mathbf{q} + \tau_c \bar{\mathbf{P}}, \\ \mathbf{u}_\perp &:= \mathbf{q}_c, & \hat{\mathbf{P}} &:= \bar{\mathbf{P}} + \alpha \mathbf{u}_\perp, \\ \mathbf{Q} &:= \mathbf{q}_c + (h - \tau_c) \hat{\mathbf{P}} & \mathbf{P} &:= \hat{\mathbf{P}} - \frac{h}{2} \nabla U(\mathbf{Q}),\end{aligned}$$

The coefficient α is chosen so that the kinetic energy is conserved through collision, thus

$$\alpha = -2 \frac{\mathbf{u}_\perp \cdot \bar{\mathbf{P}}}{\mathbf{u}_\perp \cdot \mathbf{u}_\perp}.$$

We next calculate the change in energy in a single step by inserting $\mathbf{Q}$ and $\mathbf{P}$ into the Hamiltonian and expanding around the point of collision $\mathbf{q}_c$, obtaining

$$\begin{aligned}\Delta H &\stackrel{\text{def}}{=} H(\mathbf{Q}, \mathbf{P}) - H(\mathbf{q}, \mathbf{p}) \\ &= -(h - 2\tau_c) \frac{\mathbf{q}_c^T \bar{\mathbf{P}}}{\mathbf{q}_c^T \mathbf{q}_c} \mathbf{q}_c^T \nabla U(\mathbf{q}_c) + O(h^2)\end{aligned}$$

Energy is therefore conserved only to first order through the collision unless one of the following conditions holds:

- there is no collision
- the collision occurs at the end of a timestep
- the collision occurs at the middle of the timestep, $\tau_c = h/2$,
- the directional derivative of U along the collision vector ($\mathbf{u}_\perp = \mathbf{q}_c$) vanishes, or
- the momentum vector is orthogonal to the collision vector at the point of contact.

In general, because an error of size $\mathcal{O}(h)$ occurs in each collision and there are a finite number of collisions in a fixed time interval, the total error is also $\mathcal{O}(h)$, i.e. first order.